

Nigeria CPI and Naira–USD Exchange Rate: Manager Briefing

NUTDTS816 — Time Series Analysis | MSc Data Science, NUTM | Student: Philip Adeiza Oricha (232523008)

1. Nigeria Consumer Price Index (CPI)

What the series is doing

Nigeria CPI rises persistently and increasingly steeply across 2014–2025. Growth is gradual to about 2019, accelerates from 2020, and steepens further after the Federal Government's removal of the petrol (PMS) subsidy on 29 May 2023, whose effects are visible in the data from June 2023 (NBS headline inflation rose from 22.41% in May 2023 to 22.79% in June 2023). The series is not stationary in level. It should be read as compounding growth, not fluctuation around a stable average.

What is regular, and what is not

STL decomposition assigns almost all systematic variation to trend (trend strength 0.999) and effectively none to a repeating annual cycle (seasonal strength 0.000). Monthly price movements are therefore driven by the ongoing inflationary trend rather than a predictable calendar pattern. The subsidy-removal shock is not absorbed cleanly by trend or season. The remainder (STL residual) widens sharply from 2023, showing that the decomposition treats this period as an irregular disturbance layered on top of an already-steepening trend, rather than routine variation.

Forecasting considerations

Forecasts built on raw CPI levels will be misleading. The series requires transformation and differencing before modelling. A log transform reduces curvature and variance materially, though a Box–Cox estimate of $\lambda = -0.429$ suggests some residual variance instability even after logging. Most importantly, the 2023 subsidy shock is a structural break, not noise. Models should include an explicit break term (an intervention or level-shift variable dated to mid-2023) rather than assume pre-2023 dynamics will continue unchanged. Seasonality can safely be treated as a minor, secondary feature.

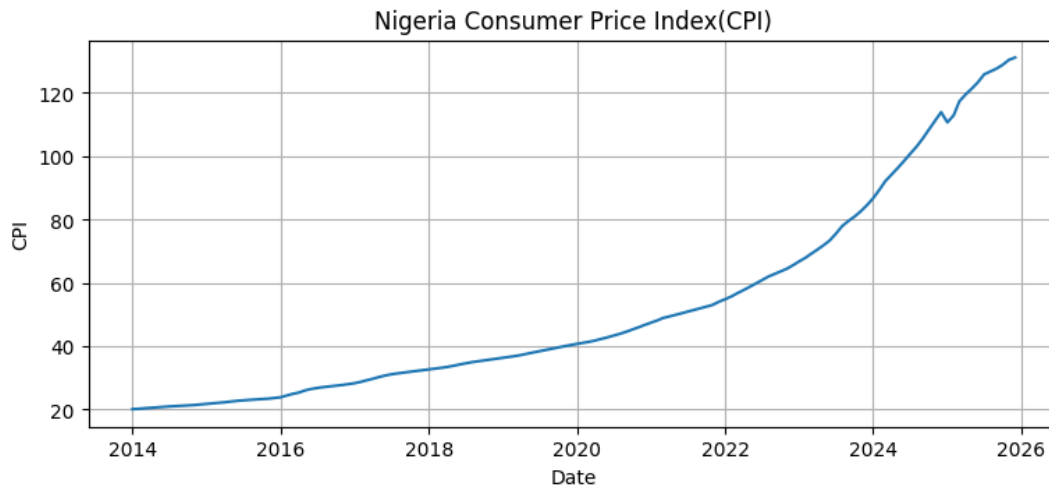


Figure 1. Nigeria CPI, January 2014–December 2025. The trajectory steepens markedly after the 2023 subsidy shock.

2. Naira–USD Exchange Rate

What the series is doing

The Naira–USD exchange rate holds a comparatively stable, lower level from 2015 through 2022, then shifts abruptly upward from 2023 into a much higher and considerably more volatile regime. This coincides with the Central Bank of Nigeria's unification of the foreign-exchange market under a willing-buyer/willing-seller framework on 14 June 2023. The rate roughly tripled within about a year of that reform and has since fluctuated at an elevated level rather than settling onto a new stable path.

What is regular, and what is not

As with CPI, STL attributes the overwhelming share of systematic movement to trend (trend strength 0.983) and almost none to a repeating annual pattern (seasonal strength 0.000). The modest wiggle STL labels 'seasonal' is better read as residual volatility than a genuine calendar effect. Classical and STL decomposition agree on the broad shape but disagree materially around 2023. Classical decomposition's fixed-window moving average lags the abrupt shift and pushes the disturbance into an inflated remainder, whereas STL's locally weighted trend adapts faster and absorbs more of the shift directly into trend. Either way, the 2023 devaluation is a level and volatility break, not a seasonal event.

Forecasting considerations

Relationships estimated on pre-2023 data should not be extrapolated across the 2023 break without adjustment. A log transform compresses scale and is useful for modelling proportional changes, but it does not remove the level shift itself (Box-Cox $\lambda = -0.682$). Forecasters should model the regime change explicitly (for example, with a dated structural-break or intervention term), and should expect and plan for heteroskedasticity (volatility clustering) in the post-2023 period rather than assuming constant variance.

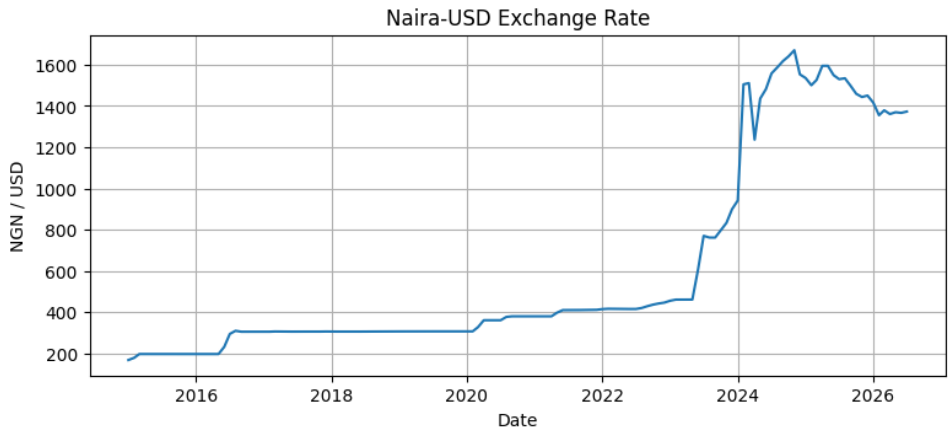


Figure 2. Naira–USD exchange rate, January 2015–mid-2026, showing the 2023 regime shift.

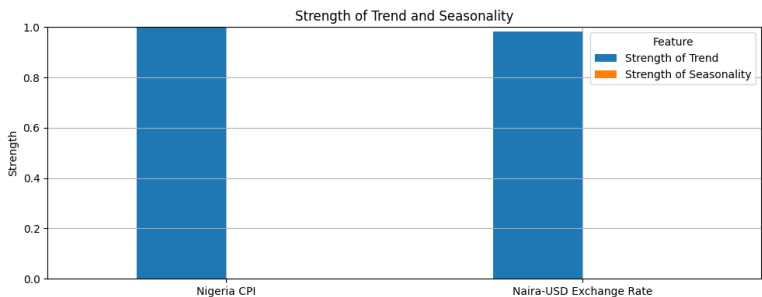


Figure 3. STL strength-of-trend and strength-of-seasonality for both series: trend dominates; seasonality is negligible in each case.

AI-use statement

Generative AI (Claude, Anthropic) was used to assist with reviewing my Python code, interpreting decomposition and diagnostic output, and improving the clarity of the written narratives. All data sourcing, analysis, results and conclusions were carried out and verified by me, and I am accountable for every line of code and every conclusion in this report and can explain any part of it orally.